SAHRA–KAVERI TWIN OBSERVATORY PROGRAMME

CONSOLIDATED TECHNICAL YEARBOOK · 2025

Site ALPHA «MIRAGE-1», Qasr Dunes margin · Site BETA «KAVERI DELTA», coastal strip

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Foreword

The 2025 yearbook is the second consolidated volume covering both observatories and the first to be compiled under the joint data-governance charter. It merges what were, until last year, two separately circulated site reports with incompatible conventions — a merger that deliberately preserves several of those inconsistencies for archival fidelity.

Readers should note that Site ALPHA records retain the DD.MM.YYYY date convention of the desert operations team, while Site BETA logs use ISO 8601. Currency figures appear in the currency of expenditure — rials, rupees or euros — with the separator conventions of the originating office. Appendix C provides the conversion basis.

«Two deserts of data — one of sand, one of rain — and a single instrument: patience.»

Programme milestones of the year include the completion of aerosol transect T-7, the first full-season operation of the cyclone moorings at BETA, and a 31 per cent reduction in diesel dependency at ALPHA. Setbacks are reported with equal candour: seventeen sand-storm shutdowns, one flooded cable trench, and the loss of sonde T7-09 to a fishing trawl.

This volume was compiled by the Programme Office from partner submissions received before 12 January 2026. Late corrections appear only in the Annex reprint and are flagged as non-authoritative.
— *The Editors*

1 · Programme Overview

The programme operates two permanently staffed observatories separated by a great-circle baseline of several thousand kilometres (Figure 1-1), exchanging calibrated aerosol, radiation and boundary-layer datasets under a common schema. Site summary cards follow; each embeds its own sub-table of commissioning data.

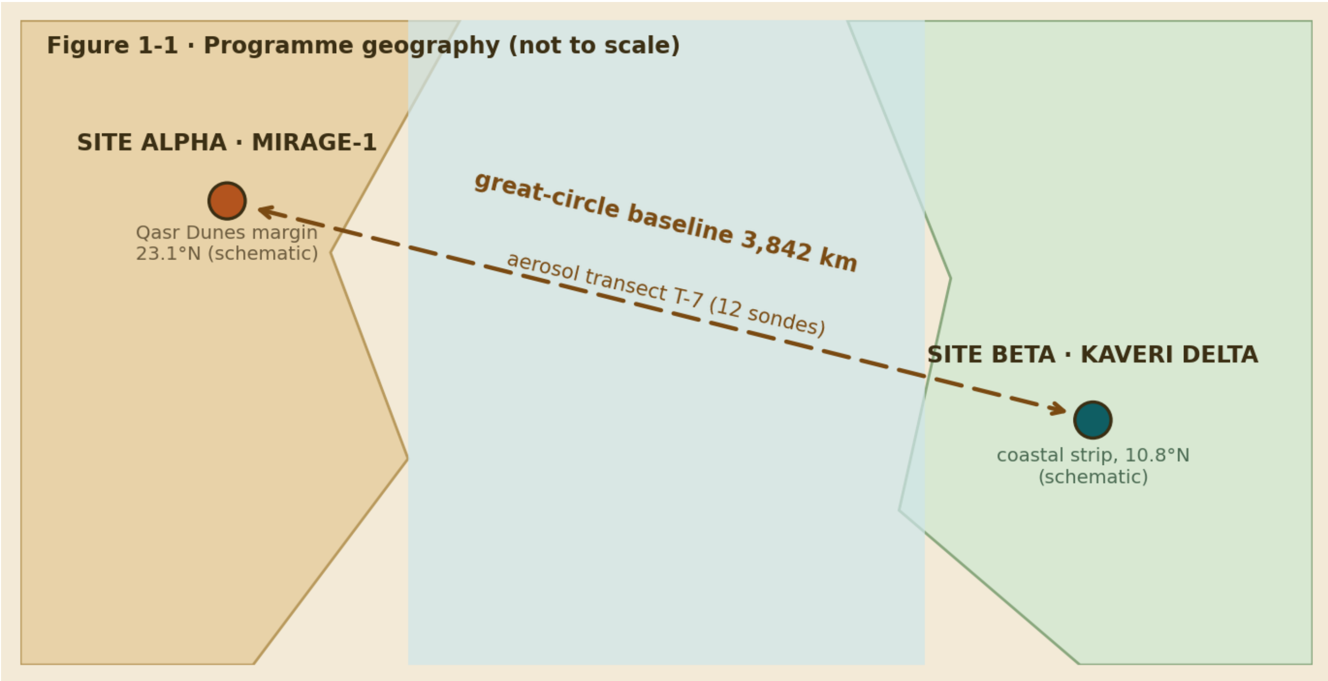

Figure 1-1. Programme geography. The baseline distance and transect designation appear only in this figure.

SITE ALPHA · desert node	
Callsign	MIRAGE-1
Elevation	612 m a.s.l.
Commissioned	November 2022
Resident crew	9
Prevailing hazard	sand storms
Grid power	none — island grid

SITE BETA · monsoon node	
Callsign	KAVERI DELTA
Elevation	4 m a.s.l.
Commissioned	March 2023
Resident crew	7
Prevailing hazard	cyclonic storms
Grid power	state feed + backup

Table 1-1. Site summary cards — an outer table whose cells each contain a nested sub-table.

2 · Instrumentation

Core instruments common to both sites are specified in Table 2-1. To conserve width, the numeric column headings are rotated vertically. Manufacturer details are given as endnotes[e1] in Appendix A.

Instrument	Site(s)	Range (km)	Rate (Hz)	Power (W)	Mass (kg)	Uptime tgt (%
Aerosol lidar VLX-9[e1]	both	15	20	410	142	95
Pyranometer-radiometer RAD-22B[e2]	both	—	1	6	2.1	98
Doppler sodar SD-4	ALPHA	0.6	0.1	220	88	90
Eddy-covariance flux tower	BETA	—	10	95	640	93
Ceilometer CL-Sigma	both	7.7	2	60	31	92

Table 2-1. Instrument envelope. The five right-hand column headings are rotated 90°.

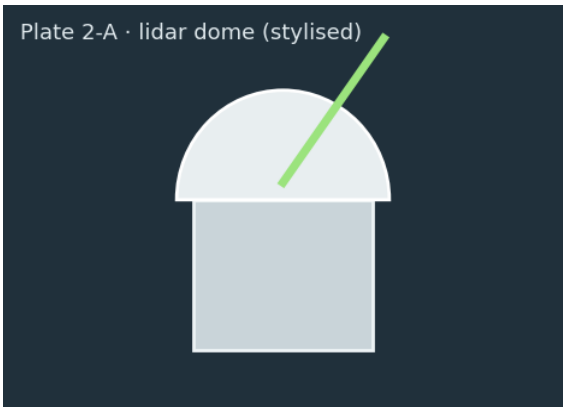

Plate 2-A. The VLX-9 lidar dome at Site ALPHA (stylised).

2.2 Site-specific instruments

Site	Instrument	Purpose	Installed
ALPHA	Saltation impact array (12 posts)	dune-margin sand transport	2023-02
	Wash-water bowser telemetry	dome cleaning logistics	2024-06
BETA	Tide & surge gauge pair	cyclone surge validation	2023-05
	Salinity string (4 nodes)	estuarine mixing	2024-11

Table 2-2. Site-specific installations (merged site cells).

3 · Desert Operations — Site ALPHA

Site ALPHA operates under the sand-storm shutdown protocol agreed with the regional partner. The authoritative statement of the protocol, as lodged in Muscat, is reproduced below in the original Arabic; no English translation is circulated.[e3]

يقع مرصد ميراج ١ على الحافة الجنوبية لكثبان القصر. عند انخفاض مدى الرؤية إلى أقل من ٨٠٠ متر يُفعَّل بروتوكول الإغلاق الرملي تلقائيًا. سُجِّل خلال عام ٢٠٢٥ سبعة عشر حدث إغلاق بمتوسط مدة أربع ساعات لكل حدث. يتولى مركز أبحاث الصحراء صيانة أجهزة قياس الغبار بموجب اتفاقية الشراكة الموقعة في مسقط.

Statement 3-1 (Arabic, right-to-left; reproduced as submitted facsimile).

3.0 Shamal season narrative

The summer shamal arrived eleven days earlier than the climatological median and persisted in three pulses of six to nine days. Station operations adapted by shifting lidar maintenance to pre-dawn windows and by pre-positioning the wash-water bowser at the dome apron. No instrument damage was recorded during any pulse, although optical throughput on the ceilometer declined 8 per cent between cleanings.

Crew rotations during the season were staged through the coastal airstrip rather than the desert track, adding four hours to each rotation but avoiding convoy exposure during peak visibility events. The protocol threshold (see Statement 3-1) was not breached during any rotation window.

3.1 Dust climatology

Monthly mean dust flux for both sites is mapped in Figure 3-2; the annual maximum at ALPHA occurred in mid-summer, coincident with the strongest shamal episodes.

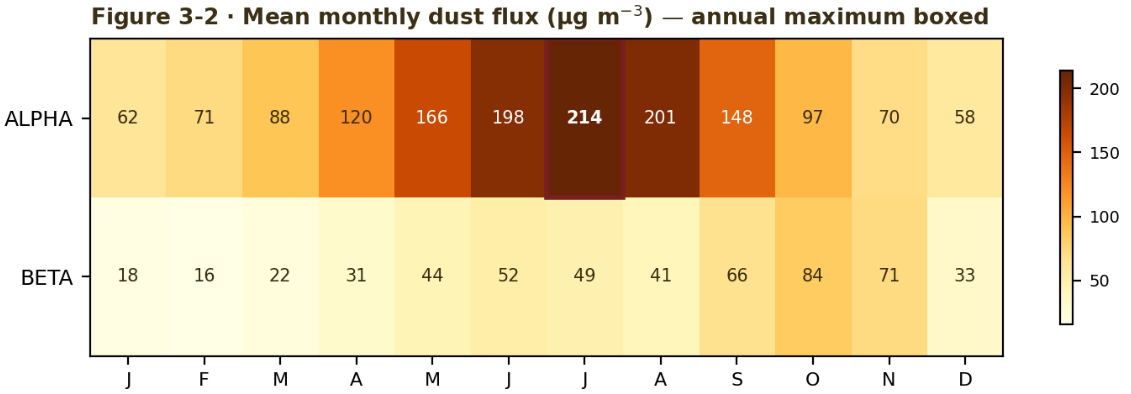

Figure 3-2. Dust-flux matrix. All values exist only in this raster.

3.2 Shutdown log

The complete FY2025 shutdown register follows. **Dates in Table 3-1 use the desert team's DD.MM.YYYY convention**; durations use the decimal comma of the originating office.

Event	Date	Visibility floor (m)	Duration (h)	Generator load shed (kWh)
S-01	12.01.2025	640	3,5	41,0
S-02	03.02.2025	710	2,0	22,5
S-03	19.02.2025	520	5,5	66,0
S-04	28.02.2025	760	1,5	18,0
S-05	14.03.2025	430	6,0	71,5
S-06	02.04.2025	690	3,0	36,0
S-07	17.04.2025	550	4,5	52,0
S-08	06.05.2025	470	5,0	61,5
S-09	23.05.2025	720	2,5	28,0
S-10	11.06.2025	390	7,5	89,0
S-11	04.07.2025	310	9,0	104,5
S-12	21.07.2025	480	4,0	47,5
S-13	09.08.2025	560	3,5	40,0
S-14	27.08.2025	610	3,0	35,5
S-15	15.09.2025	700	2,5	27,0
S-16	08.10.2025	740	2,0	23,5
S-17	19.11.2025	770	1,5	17,0

Table 3-1. Sand-storm shutdown register (17 events). Decimal commas per originating-office convention.

For the avoidance of doubt: the evacuation *drill* of 7 April was a scheduled exercise and is not a shutdown event; it therefore does not appear in Table 3-1.

4 · Monsoon Operations — Site BETA

Site BETA's marine infrastructure statement, submitted by the coastal operations team, is reproduced in the original Tamil; as with Statement 3-1, no translation is circulated.

> காவேரி டெல்டா நிலையத்தின் கடல் நங்கூரங்கள் மணிக்கு 180 கிலோமீட்டர் வேகக் காற்றைத் தாங்கும் வகையில் வடிவமைக்கப்பட்டுள்ளன. பருவமழைக் காலத்தில் தரவுப் பெட்டகங்கள் வாரம் இருமுறை நாகப்பட்டினம் மீனவர் கூட்டுறவு சங்கப் படகுகள் மூலம் கரைக்கு அனுப்பப்படுகின்றன. மின்தடை நேரங்களில் நிலையம் தானியங்கு முறையில் குறைந்த ஆற்றல் நிலைக்கு மாறும்.

Statement 4-1 (Tamil; reproduced as submitted facsimile).

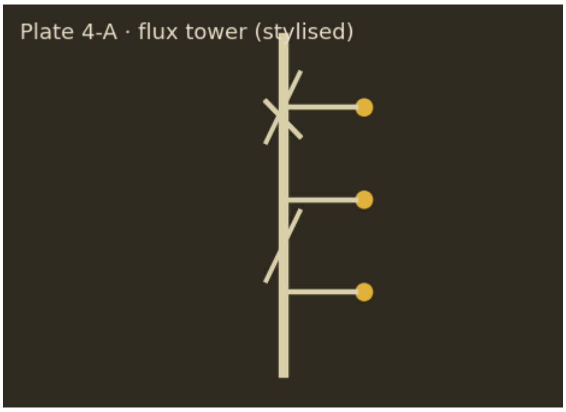

Plate 4-A. BETA eddy-covariance flux tower (stylised).

4.1 Mooring array

Mooring	Position (rel.)	Depth (m)	Deployed	Recovered	Condition
M-1	2.1 km NE	18	2025-05-02	2025-11-20	good
M-2	3.4 km E	26	2025-05-02	2025-11-21	biofouled
M-3	5.0 km ESE	31	2025-05-03	—	lost — trawl strike
M-4	1.2 km S	12	2025-05-03	2025-11-20	good

Table 4-1. Cyclone-season mooring campaign. BETA logs use ISO 8601 dates throughout.

4.2 Monsoon data courier

During the north-east monsoon the terrestrial backhaul degrades below the exchange gateway minimum, and physical media couriers take over per Statement 4-1. The season schedule ran as follows:

Window	Frequency	Carrier	Media
Oct 15 – Nov 30	Twice weekly	Cooperative boats (Nagapattinam)	2 × 8 TB cases
Dec 01 – Jan 15	Weekly	Cooperative boats (Nagapattinam)	2 × 8 TB cases
Jan 16 – Feb 28	Fortnightly	Programme vehicle	1 × 8 TB case

Table 4-2. Courier schedule. The peak frequency corroborates Statement 4-1 (Tamil).

5 · Cross-Site Science

5.1 The June dust intrusion

The June maximum columnar aerosol optical depth at BETA reached **0.82** during the dust intrusion of 14 June, the highest single-day value of the record (Figure 5-1).

The June median columnar aerosol optical depth at BETA remained **0.28** outside the intrusion window, the lowest monthly median of the record.

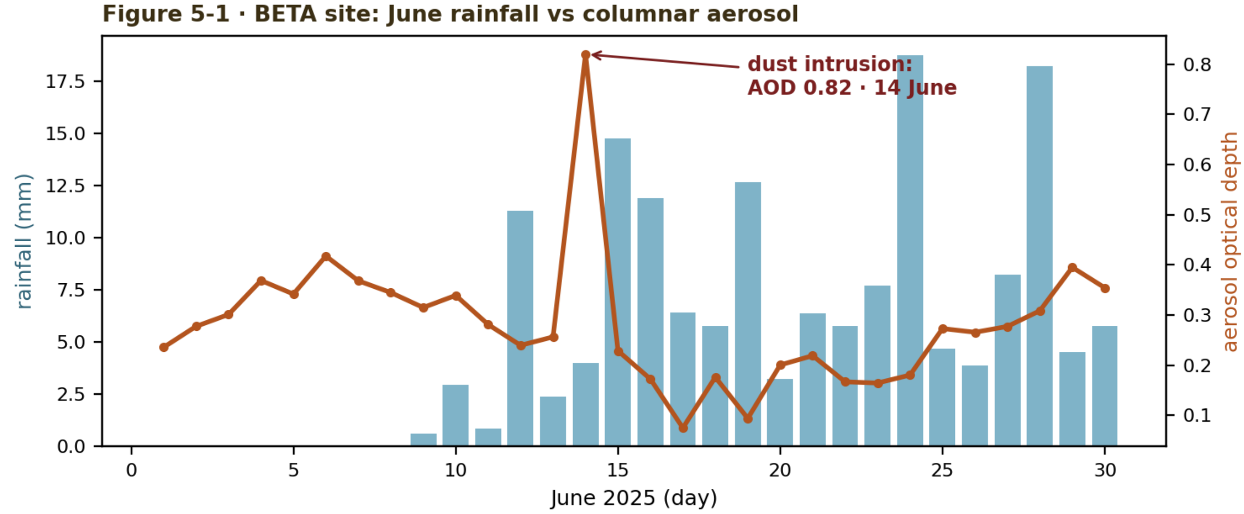

Figure 5-1. Dual-axis series: rainfall (left) vs AOD (right). The annotated peak exists only in the raster.

Ångström exponents were derived as $\alpha = -\ln(\tau_1/\tau_2) / \ln(\lambda_1/\lambda_2)$, with $\lambda_1 = 440$ nm and $\lambda_2 = 870$ nm.[e4]

5.2 Instrument availability

Pooled availability by instrument class is charted in Figure 5-2. The lidar figure of 97.2 per cent is the authoritative value for FY2025 and is restated in the Annex reprint.

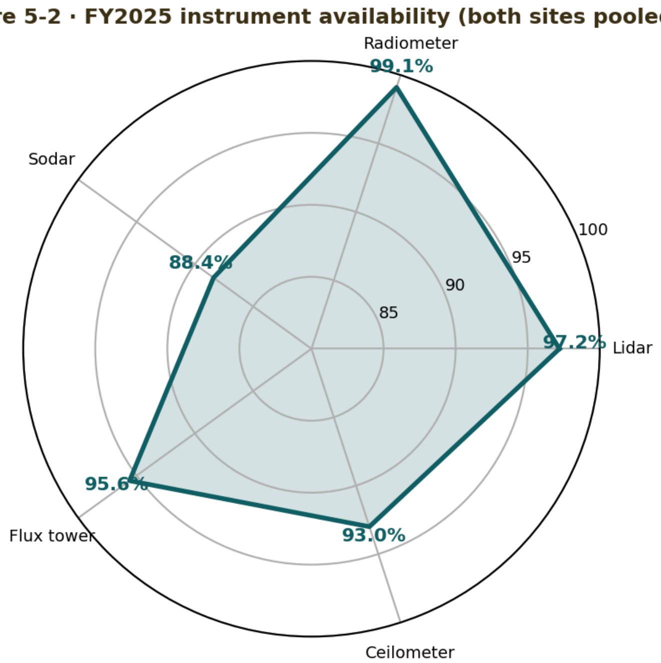

Figure 5-2. Availability radar. Percentages appear only in the chart.

5.3 Transect T-7 sonde campaign

Twelve driftsondes were released along transect T-7 between 02 and 19 October. Sonde T7-09 was lost to a fishing trawl on descent (log entry L-017); all other packages were recovered. Trajectories are archived under campaign key SK-T7-2025.

Sonde	Release (UTC)	Apogee (hPa)	Drift (km)	Recovery
T7-01	02 Oct 04:12	62	1,940	beach — intact
T7-02	03 Oct 04:20	61	2,120	vessel — intact
T7-03	05 Oct 04:05	64	1,780	beach — intact

Sonde	Release (UTC)	Apogee (hPa)	Drift (km)	Recovery
T7-04	06 Oct 04:31	60	2,410	vessel — antenna bent
T7-05	08 Oct 04:18	63	2,050	beach — intact
T7-06	09 Oct 04:09	59	2,660	vessel — intact
T7-07	11 Oct 04:26	62	2,230	beach — intact
T7-08	13 Oct 04:14	61	2,090	vessel — intact
T7-09	14 Oct 04:22	63	—	lost — trawl strike
T7-10	16 Oct 04:07	60	2,480	vessel — intact
T7-11	17 Oct 04:29	62	2,150	beach — intact
T7-12	19 Oct 04:11	61	1,990	beach — intact

Table 5-1. T-7 driftsonde campaign. Mean drift of recovered sondes: 2,172 km.

5.4 Cross-site intercomparison statistics

Variable	ALPHA			BETA		
	mean	σ	n	mean	σ	n
AOD 550 nm	0.41	0.19	341	0.33	0.15	352
Ångström α (440/870)	0.36	0.22	341	0.94	0.31	352
PBL height (m, noon)	2,140	480	298	870	260	305
Global irradiance (W m^{-2}, noon)	958	112	360	842	189	358

Table 5-2. Annual statistics; two-level header groups three sub-columns per site.

6 · Power & Logistics

Across the year **38,400 litres** of diesel were consumed at ALPHA — a figure the energy dashboard disputes; see Table 6-1, whose quarterly sub-tables are nested inside the outer summary and which totals **34.800** litres (note the originating office's thousands separator). Reconciliation is pending.[e5]

<table>
<tr><th>Quarter</th><th>Line items (nested)</th><th>Diesel (L)</th></tr>
<tr><td>Q1</td><td><table><tr><td>Generator set A</td><td>4.980</td></tr><tr><td>Generator set B</td><td>3.310</td></tr><tr><td>Vehicles</td><td>1.240</td></tr></table></td><td>9.530</td></tr>
<tr><td>Q2</td><td><table><tr><td>Generator set A</td><td>4.120</td></tr><tr><td>Generator set B</td><td>2.980</td></tr><tr><td>Vehicles</td><td>1.400</td></tr></table></td><td>8.500</td></tr>
<tr><td>Q3</td><td><table><tr><td>Generator set A</td><td>4.610</td></tr><tr><td>Generator set B</td><td>3.150</td></tr><tr><td>Vehicles</td><td>1.140</td></tr></table></td><td>8.900</td></tr>
<tr><td>Q4</td><td><table><tr><td>Generator set A</td><td>3.720</td></tr><tr><td>Generator set B</td><td>2.870</td></tr><tr><td>Vehicles</td><td>1.280</td></tr></table></td><td>7.870</td></tr>
<tr><td>Total</td><td></td><td>34.800</td></tr>
</table>

Table 6-1. ALPHA diesel consumption. Each quarter's cell embeds a nested line-item table; values use period as thousands separator (34.800 = thirty-four thousand eight hundred).

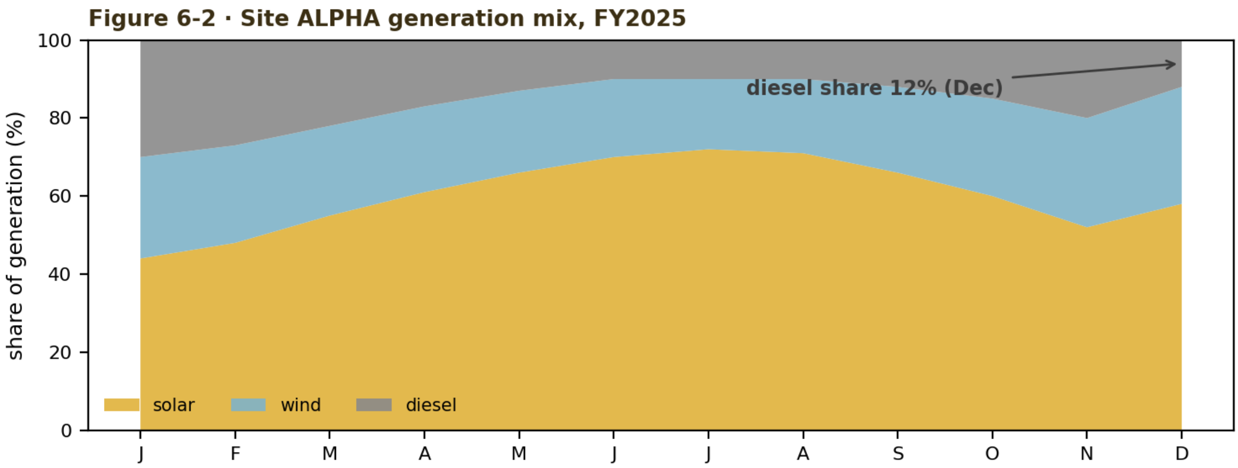

Figure 6-2. Generation mix. The December diesel share is annotated only in the raster.

6.1 Cross-currency procurement

Item	Office	Amount (as booked)	EUR equivalent
Mooring hardware	BETA (INR)	₹ 42,15,600	€ 46.310,25
Lidar window assemblies	ALPHA (OMR)	OMR 8 412,500	€ 20.104,80
Sonde consignment T-7	Programme (EUR)	€ 61.900,00	€ 61.900,00

Table 6-2. Procurement in three locale conventions: Indian lakh grouping (42,15,600), Omani space grouping with three decimals, and European decimal comma.

7 · Data Governance

The exchange gateway is configured as follows (schema v2, YAML):


```yaml
gateway:
  name: sk-exchange
  version: 2.4
  sync_window_utc: "02:00-04:30"
  max_payload_mb: 96
  checksum: xxh3
  retry: {attempts: 5, backoff: exponential}
  quarantine_days: 14
```


Gateway administrative access code: [redacted] (redacted in the distribution copy under governance directive DG-9 §3; the code is held by the Programme Office and is not recoverable from this document).

Data class	Embargo	Raw retention	Steward
Lidar profiles	none	permanent	Programme Office
All-sky imagery	90 days	8 years	site teams
Sonde profiles (T-series)	30 days	permanent	Programme Office
Campaign notebooks	365 days	15 years	originating PI

Table 7-1. Embargo and retention schedule under the joint charter.

8 · Personnel Exchange Matrix

Cross-postings for the year are charted below. **Note:** in the distribution PDF this page is intentionally stored with a 90° rotation flag to exercise reader compliance.

Staff member	Home	J	F	M	A	M	J	J	A	S	O	N	D
A. Al-Harthy	ALPHA												
R. Subramani	BETA												
D. Okafor	Programme												
M. Steiner	ALPHA												
P. Chandran	BETA												
L. Ivanova	Programme												
K. Al-Busaidi	ALPHA												
S. Meenakshi	BETA												

Table 8-1. Exchange Gantt: teal = posted to BETA, sand = posted to ALPHA, green = programme office rotation. Bars are colour-fill only; no text marks the cells.

Exchange postings totalled 41 person-months, up from 28 in 2024. The longest single posting (D. Okafor, six months) supported the T-7 sonde campaign.

9 · Outreach & Publications

Programme outputs for the year, in the yearbook citation style (hanging indents):

[1] Al-Harthy, A., Steiner, M. & Okafor, D. (2025). Shamal-phase locking of dust flux maxima at a desert margin observatory. *J. Fict. Atmos. Sci.*, 12(4), 411–428.

[2] Subramani, R. & Meenakshi, S. (2025). Cyclone-season mooring survivability in shallow deltaic waters. *Fict. Ocean Eng.*, 8(2), 77–94.

[3] Okafor, D., Ivanova, L., Chandran, P. & Al-Busaidi, K. (2025). The T-7 driftsonde transect: first results. *Proc. Fict. Aerosol Conf.*, 233–241.

[4] Steiner, M. (2025). A two-site closure test for the Ångström exponent under mixed dust–marine aerosol. *Fict. Geophys. Lett.*, 3, e2025-0187.

[5] Chandran, P. & Subramani, R. (2025). Physical-media data logistics for monsoon-isolated stations. *Fict. Data Pract.*, 2(1), 9–17.

[6] Ivanova, L. (2025). Yearbook harmonisation of heterogeneous site conventions: a cautionary study. *Fict. Res. Infrastr. Rev.*, 5, 140–151.

Publication [4] provides the derivation underlying Section 5.1; publication [5] documents the courier system of Table 4-2.

10 · Risk Register (extract)

ID	Risk	Likelihood	Impact	Rating	Owner
R-03	Trawl damage to moorings/sondes	High	Medium	AMBER	BETA ops
R-07	Extended shamal beyond fuel reserve	Low	High	AMBER	ALPHA ops
R-11	Gateway credential compromise	Low	Severe	RED	Programme Office
R-14	Courier media loss at sea	Medium	Low	GREEN	BETA ops
R-18	Single-source lidar spares	Medium	High	RED	Programme Office
R-22	Visa delays for exchanges	Medium	Low	GREEN	Programme Office

Table 10-1. Register extract. The rating column encodes severity by cell colour as well as label.

CALIBRATION CERTIFICATE

No. VC-25-4471

INSTRUMENT	Pyranometer-Radiometer RAD-22B
SERIAL	SK-BETA-0071
METHOD	Side-by-side vs. reference WRR-traceable cavity
CALIBRATION FACTOR	1.0342 (k=2 uncertainty ±0.6 %)
AMBIENT	24.8 °C / 41 %RH
DATE OF ISSUE	03.11.2025
NEXT CALIBRATION DUE	03.11.2026
APPROVED	Dr. F. Marek — Laboratory Head

VESPERA
OPTIK
JENA

This facsimile is valid without signature per QM-7 §4.

11 · Joint Steering Meeting — Minutes (extract)

SK-JSM-2025-04 · held by videolink 21 November 2025, 09:00 UTC. Present: I. Halonen (chair), A. Al-Harthy, R. Subramani, L. Ivanova, D. Okafor, M. Steiner (minutes). Apologies: P. Chandran.

Item 3 — diesel reconciliation. Finance restated the dashboard figure and the yearbook figure side by side; the working group (endnote e5) was directed to report by the Q2-2026 plenary. The chair noted that no procurement action is blocked meanwhile.

Item 5 — mooring M-3 loss. Insurance recovery of €14.200,00 was confirmed. A replacement mooring will be fabricated locally and deployed before the 2026 season.

Item 7 — gateway credential rotation. Following risk R-11, credentials will rotate quarterly from January 2026; the access code remains restricted under DG-9 §3.

Action	Owner	Due	Status at issue
A-41: Deliver diesel reconciliation report	Finance WG	Q2-2026 plenary	open
A-42: Fabricate & deploy replacement mooring M-3R	BETA ops	2026-04-30	open
A-43: Implement quarterly credential rotation	Programme Office	2026-01-15	open
A-44: Second-source lidar spares study (risk R-18)	M. Steiner	2026-03-31	open

Table 11-1. Actions issued at SK-JSM-2025-04.

12 · Station Diaries (extracts)

ALPHA, 4 July. — *Shut down at 09:40 local when the ridge vanished. Visibility meter read 310 before we lost the mast camera to static. Nine hours in the mess playing carrom; generator shed 104.5 units before the wind turned. Longest event of my three seasons. — K.A-B.*

BETA, 14 June. — *Sky the colour of unpolished brass all afternoon. Sun photometer needle went places I have not seen outside the textbook; logged the column at 0.82 at half past two and again at three. Fisherfolk stayed in; the harbourmaster called twice to check the flags. — S.M.*

BETA, 14 October. — *Trawler out of the northern grounds took T7-09 on descent, clean through the chute lines. Skipper radioed apologetic; brought us pomfret by way of settlement. Insurance paperwork weighs more than the sonde did. — R.S.*

Diary extracts are informal and unedited; where they conflict with tables, tables govern.

Annex R · Reprint of §5.2 as circulated to partners

The following text was circulated to partners on 19 December 2025, before final QC. It is reproduced verbatim for archival completeness and is **not authoritative**.

5.2 Instrument availability (as circulated)

Pooled availability by instrument class is charted in Figure 5-2. The lidar figure of **97.8 per cent** is the authoritative value for FY2025 and is restated in the Annex reprint.

Radiometer availability of 99.1 per cent, sodar 88.4, flux tower 95.6 and ceilometer 93.0 complete the pooled set for the reporting year.

Erratum note: partners should rely on the body text of Section 5.2, not this reprint, where the two disagree.

Appendix A · Endnotes

e1 The VLX-9 aerosol lidar is manufactured by Vespera Optik GmbH of Jena (fictional); units carry a five-year emitter warranty.

e2 RAD-22B radiometers are recalibrated annually; the FY2025 certificate for the BETA unit is reproduced as the facsimile page preceding Annex R.

e3 The Arabic statement was lodged with the partnership registry in Muscat on 9 February 2025 under file P-114/25.

e4 Wavelength pair follows the programme convention adopted at the 2023 plenary; earlier site reports used 500/870 nm and are not directly comparable.

e5 The reconciliation working group attributes the diesel discrepancy to bunkered fuel transferred to the drilling contractor and expects closure in Q2 2026.

Appendix B · Consolidated Station Log (extract)

Sixty representative entries. ALPHA rows keep DD.MM.YYYY dates; BETA rows use ISO 8601 — the mixture is preserved deliberately (see Foreword).

Entry	Site	Date	Activity	Operator	Flag
L-001	ALPHA	25.05.2025	routine calibration	AH	
L-002	BETA	2025-10-10	mooring check	DO	
L-003	ALPHA	17.06.2025	visitor day	KB	
L-004	BETA	2025-05-15	visitor day	RS	
L-005	ALPHA	20.08.2025	filter change	AH	
L-006	BETA	2025-02-23	mooring check	DO	
L-007	ALPHA	01.09.2025	sonde launch	LV	
L-008	BETA	2025-01-07	spares inventory	RS	
L-009	ALPHA	12.11.2025	spares inventory	KB	!
L-010	BETA	2025-04-24	spares inventory	SM	
L-011	ALPHA	25.06.2025	sonde launch	MS	
L-012	BETA	2025-10-03	data gap logged	DO	
L-013	ALPHA	23.08.2025	data gap logged	LV	
L-014	BETA	2025-03-26	antenna alignment	PC	
L-015	ALPHA	11.11.2025	spares inventory	KB	
L-016	BETA	2025-02-03	mooring check	RS	
L-017	ALPHA	02.10.2025	sonde launch — T7-09 lost to trawl	MS	!
L-018	BETA	2025-10-09	filter change	SM	
L-019	ALPHA	12.05.2025	software patch	MS	
L-020	BETA	2025-09-07	data gap logged	SM	
L-021	ALPHA	04.10.2025	software patch	AH	
L-022	BETA	2025-04-11	spares inventory	PC	
L-023	ALPHA	04.07.2025	software patch	KB	
L-024	BETA	2025-01-02	mooring check	DO	
L-025	ALPHA	01.03.2025	routine calibration	KB	
L-026	BETA	2025-04-26	antenna alignment	RS	
L-027	ALPHA	15.12.2025	sonde launch	AH	
L-028	BETA	2025-01-26	software patch	DO	
L-029	ALPHA	07.01.2025	generator service	AH	
L-030	BETA	2025-08-18	software patch	PC	!

Entry	Site	Date	Activity	Operator	Flag
L-031	ALPHA	10.04.2025	filter change	AH	
L-032	BETA	2025-09-17	antenna alignment	PC	
L-033	ALPHA	18.04.2025	filter change	MS	
L-034	BETA	2025-07-06	software patch	RS	
L-035	ALPHA	25.06.2025	data gap logged	LV	
L-036	BETA	2025-10-23	filter change	DO	
L-037	ALPHA	23.11.2025	antenna alignment	KB	
L-038	BETA	2025-05-14	generator service	DO	
L-039	ALPHA	25.09.2025	generator service	AH	
L-040	BETA	2025-02-01	sonde launch	PC	
L-041	ALPHA	18.12.2025	visitor day	LV	
L-042	BETA	2025-03-22	sonde launch	RS	
L-043	ALPHA	18.12.2025	visitor day	AH	
L-044	BETA	2025-09-16	generator service	DO	
L-045	ALPHA	01.06.2025	routine calibration	LV	
L-046	BETA	2025-03-24	visitor day	DO	
L-047	ALPHA	26.11.2025	data gap logged	AH	
L-048	BETA	2025-11-15	filter change	DO	
L-049	ALPHA	14.12.2025	software patch	MS	
L-050	BETA	2025-10-23	antenna alignment	PC	
L-051	ALPHA	18.05.2025	filter change	MS	
L-052	BETA	2025-04-02	mooring check	DO	
L-053	ALPHA	25.01.2025	spares inventory	LV	
L-054	BETA	2025-07-10	mooring check	SM	
L-055	ALPHA	12.05.2025	sonde launch	MS	
L-056	BETA	2025-03-13	mooring check	RS	
L-057	ALPHA	27.11.2025	antenna alignment	LV	
L-058	BETA	2025-05-18	filter change	SM	
L-059	ALPHA	13.04.2025	data gap logged	MS	
L-060	BETA	2025-09-12	spares inventory	RS	

Table B-1. Consolidated log extract; exclamation flags mark incident-linked entries.

Appendix C · Conversion Basis & Colophon

Pair	Rate used	Basis date
OMR → EUR	2.3901	2025-12-31
INR → EUR	0.010986	2025-12-31

Table C-1. Year-end conversion basis.

Appendix D · Travel Authorisation (specimen form)

A completed cross-posting travel authorisation is reproduced in native layout.

Authorisation No.	TA-2025-118
Traveller	D. Okafor — Programme Office
Posting	Site BETA, T-7 campaign support
Period	01 May 2025 – 31 October 2025 (6 months)
Routing	HQ → Chennai → Nagapattinam → Site BETA

Per-diem class	B2 — coastal field rate		
Advance	€ 3.850,00 (booked to programme code SK-73)		
Approvals	Site lead	Programme office	Finance
	☑ A.A-H. 12.03.2025	☑ L.I. 2025-03-14	☐ pending

Form SK-D-04. Note the mixed date conventions inside the approvals row — deliberate.

Appendix E · Acronyms

AOD — aerosol optical depth

DG-9 — data-governance directive 9

EC — eddy covariance

ISO 8601 — international date standard (BETA logs)

OMR — Omani rial

INR — Indian rupee

QC — quality control

SK — SAHRA–KAVERI programme

T-7 — aerosol transect seven

UTC — coordinated universal time

WRR — world radiometric reference

YB — yearbook

Colophon. Synthetic stress-test artefact #2 for retrieval-augmented generation pipelines. Novel features relative to artefact #1: three-column layout, rotated text inside table cells, nested tables, right-to-left Arabic and Tamil facsimiles, a page stored with a 90° rotation flag, a near-duplicate annex page differing in a single figure, mixed date and number locales, endnotes, a redacted access code, an invisible extraction canary, and a metadata keyword canary. All entities are fictional; the companion answer key enumerates every trap.